Food Access in North Carolina

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Your written report goes here!

> **!** Important
>
> Before you submit, make sure your code chunks are turned off with `echo: false` and there are no warnings or messages with `warning: false` and `message: false` in the YAML.

Introduction

There are 18.8 million Americans living in food deserts today — areas with limited access to affordable and nutritious food. The USDA has defined food deserts to be census tracts that have a poverty rate of 20% or greater (or median family income that is less than 80% of statewide family income) and where at least 33% of the population (or at least 500 people) live more than 1 mile from the supermarket in an urban area/ 10 miles in a rural area. Food deserts are created not only due to distance from healthy food sources but also due to structural decisions that have historically limited healthy food access for minority communities. Understanding this issue requires examining how redlining policies institutionalized neighborhood disinvestment, disproportionately exposing low-income and minority communities to less healthy food environments. These disparities carry serious consequences: food deserts are linked to worse health outcomes, including shorter life expectancy and a higher risk for adverse cardiovascular events. Given these health risks and the disproportionate racial impact of food deserts, it is critical to evaluate what factors contribute to their persistence and how current policy can work toward reversing years of racial residential segregation.

This issue is particularly relevant to us as Duke students, as 9 of the top 10 states with the greatest share of residents living in food deserts are in the South. This proximity makes understanding the structural drivers of food deserts a matter of community concern. Thus, I seek to answer the following question: in what ways do the poverty rate of a census tract, the number of households receiving SNAP benefits, the number of households without a vehicle,

median family income, and urban/rural classification influence whether a census tract is a food desert (classified as a low-income, low-access area) in North Carolina? I will use the USDA's 2019 Food Research Atlas dataset to investigate this question and illuminate patterns in food access in North Carolina

Since this dataset is substantially large with more than 72,000 observations, fitting a model on the data could result in some predictors with extremely low p values, indicating significant impacts on the response variable, though it may not truly be the case. In order to address this limitation, I will focus specifically on North Carolina to make this analysis more relevant to our positioning as residents of the state. Though filtering the data addresses the issue of having too many observations skew the p-values, observations in the dataset could be dependent due to the fact that neighboring census tracts will likely have similar measures for variables in the dataset. To account for this, I will take a random sample of 20% of the filtered, North Carolina-specific dataset. Finally, there is substantial missing data in variables that track specific numbers, like amount of people living in a low access, low income area (`LALOWI1_10`). Thus, I will use the `LILATracts_1AND10` response variable, which is an indicator variable of what `LALOWI1_10` measures. Beyond that, if some variables have only a few missing values, I will delete those observations from the data. Therefore, I also acknowledge that this data may not be entirely representative of the entire state, especially if those null values were concentrated in a specific area. Finally, in working with health and social science data, I am also cognizant of confounding being a limitation. This dataset does not provide a comprehensive account of all the factors that impact whether a census tract is a food desert since other variables can influence this measure/ the measure of other predictors. For instance, the `TractHUNV` predictor measures the number of housing units without a vehicle in a certain tract. Though the absence of a vehicle can make healthy food inaccessible, and therefore contribute to the categorization of an area as a food desert, the variable does not give a complete picture of accessibility — especially since access to public transportation is not considered. I will remain mindful of these limitations and incorporate them into my analysis to ensure that my interpretations and conclusions are statistically sound.

Exploratory Data Analysis

Data description

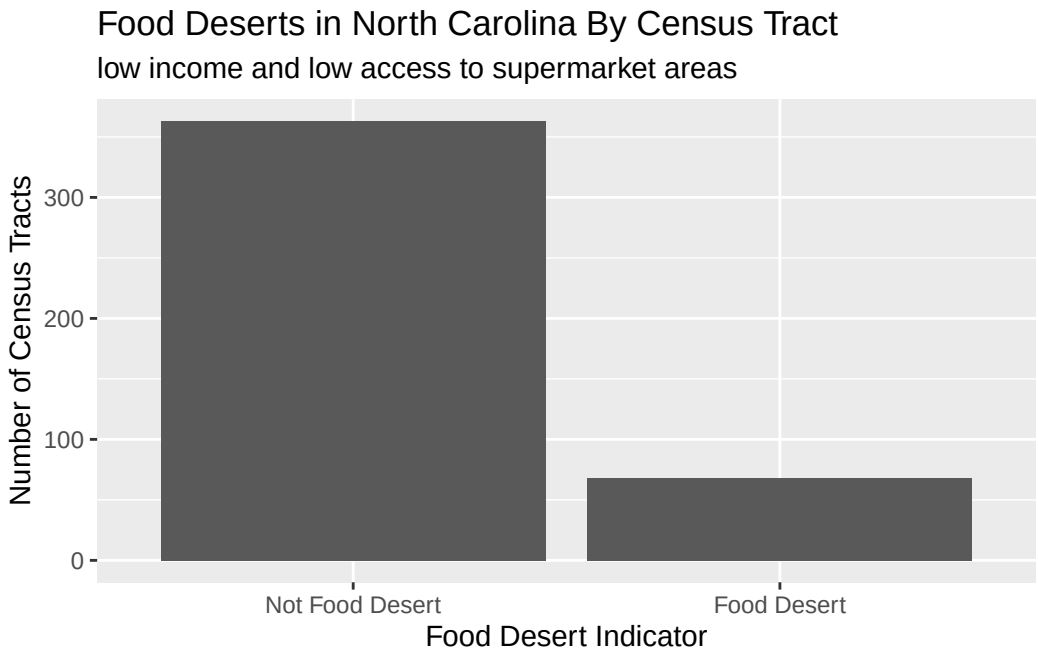
I will analyze the 2019 Food Access Research Atlas dataset from the USDA Economic Research Service. It provides an overview of food access indicators in the US by measuring variables like supermarket accessibility, race, poverty rate, and median income by census tract. This dataset is a combination of data collected by the 2010 Decennial Census, the 2014-18 American Community Survey (ACS), and two 2019 lists of supermarkets, one being from Trade Dimensions TDLinx. The 2010 Census provided population data relating to age, race, income, and poverty rate. The 2014-18 ACS contributed information about income, vehicle availability, and SNAP participation. Finally, two lists of grocery stores were combined to produce a list of stores that

have nutritious and affordable food. This data was then used to compute the variables that communicate the distance that one lives from a grocery store.

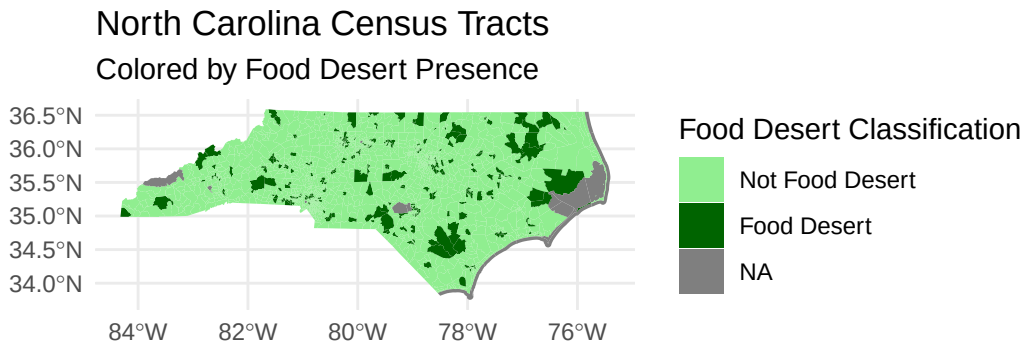
Key Variables

Each observation in the data represents an individual census tract in a specific county and state, uniquely identified by the `CensusTract` variable. In understanding factors that contribute to a census tract's categorization as low-income and low-access (or being a food desert), I will focus on the `LILATracts_1And10` response variable and the following predictor variables: `Urban` , `TractSNAP` , `TractHUNV` , `PovertyRate` , `MedianFamilyIncome` . `Urban` is an indicator variable that classifies a census tract as urban or not using the official definition by the US Census (urban area has more than 2,500 people). This predictor will contribute to understanding whether food deserts vary by urbanicity, which is helpful in understanding where policymakers should direct resources to address such issues. Next, `TractSNAP` will detail the number of people within a census tract receiving SNAP benefits. The rationale behind including this predictor is that communities with more individuals receiving SNAP benefits are also likely to be underresourced, characteristic of food deserts. `TractHUNV` provides information on access to a vehicle, which could illuminate the low access to grocery stores aspect of the food desert definition. Finally, `MedianFamilyIncome` and `Urban` will both be used to determine the general economic state of families in a census tract. Since having these two variables could present issues of multicollinearity in the model, I will check the variance inflation factor to make a decision on keeping the predictors and which one. Finally, `LILATracts_1And10` is an indicator variable which tracks if a census tract is low income and low access, measured at 1 mile for urban areas and 10 miles for rural areas. This fits the USDA's definiton of a food desert, so `LILATracts_1And10` will also serve as an abstraction for whether a census tract is classified as a food desert.

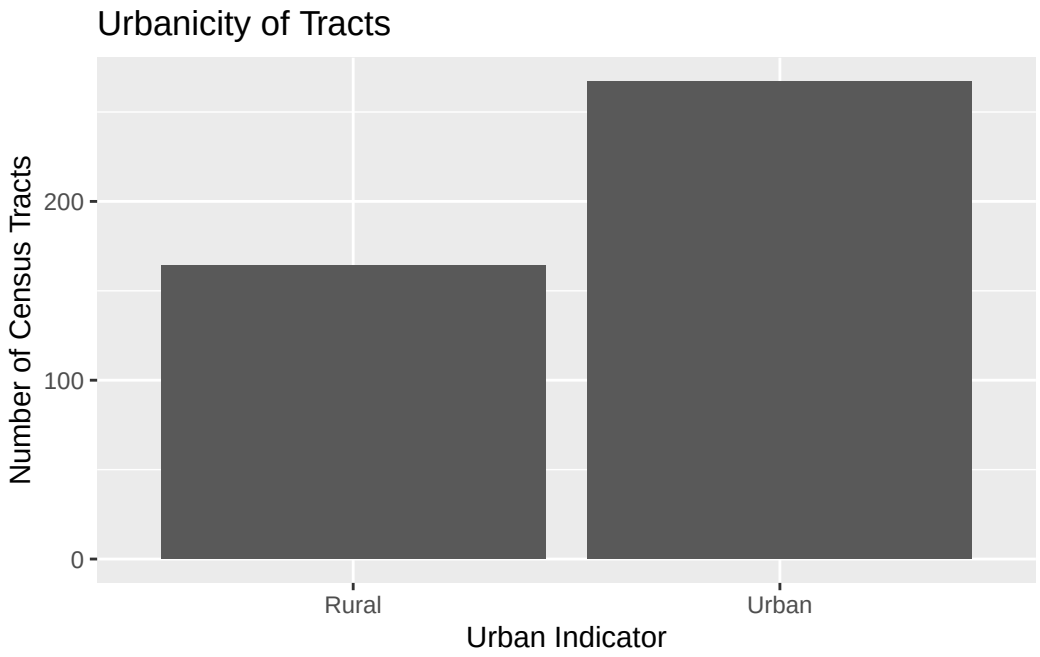
Univariate EDA

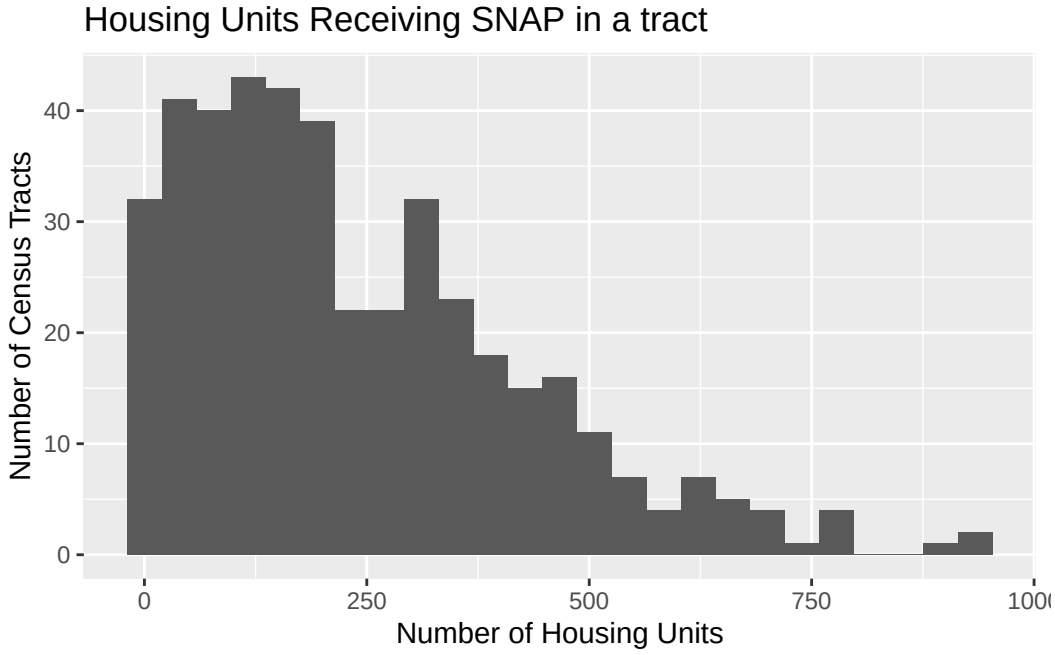


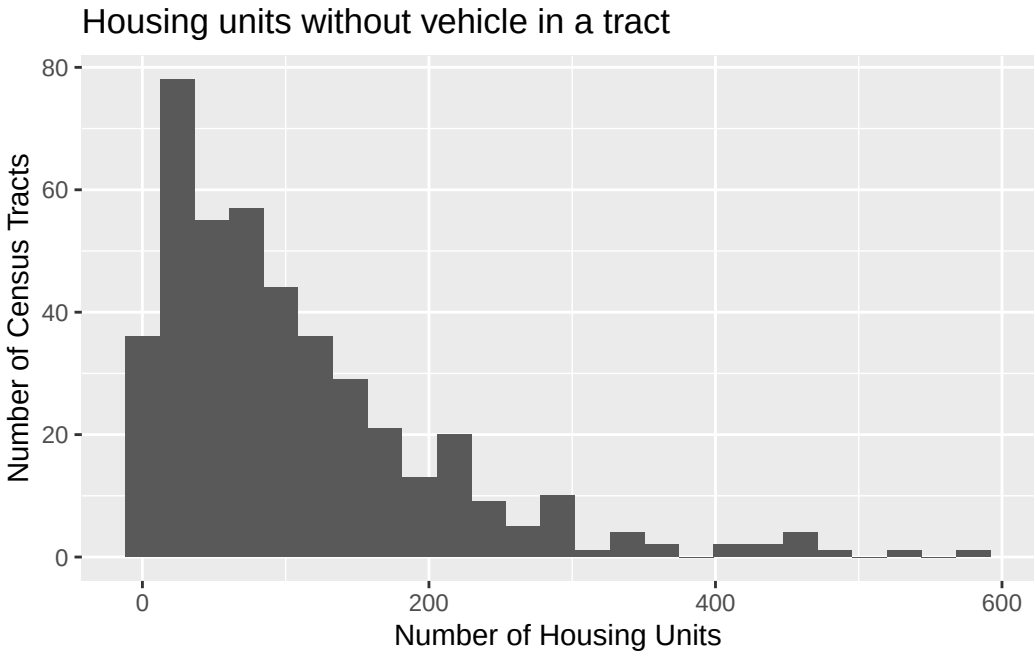

```
# A tibble: 2 x 3
  LILATracts_1And10     n  prop
  <fct>             <int> <dbl>
1 Not Food Desert     363 0.842
2 Food Desert          68 0.158
```

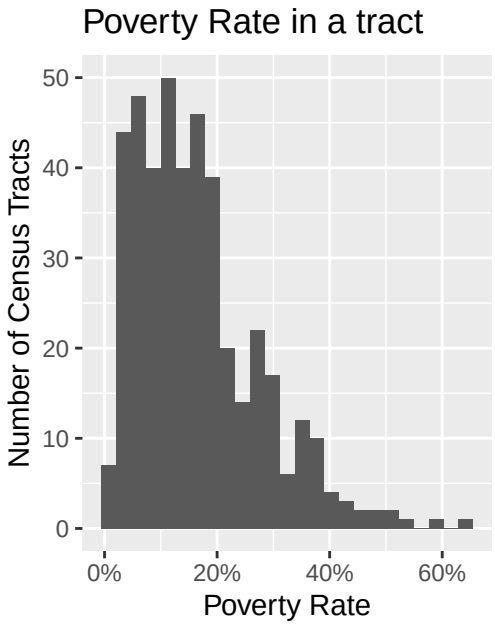


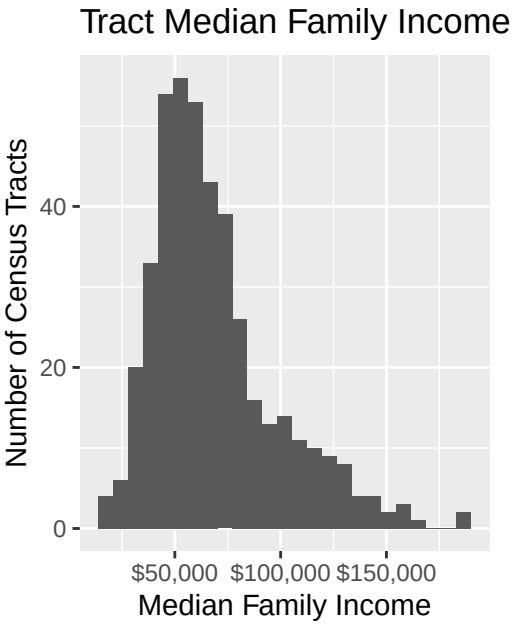
Observing the bar chart and the table about the proportion of tracts within the sample categorized as food deserts illuminates how the majority of census tracts in North Carolina (around 84%) are not food deserts. Furthermore, the map of food deserts across the state shows how food deserts are concentrated in small areas and do not simply exist at the census tract level. Noticing this is important for understanding how food deserts may not be very common, but they are evident and harmful for numerous communities across North Carolina. Finally, since `LILATracts_1And10` is mostly 0s, this suggests that the model may not predict tracts that are food deserts as well as it is able to predict tracts that are not food deserts, simply due to the amount of data it has for training. During inference, I will revisit this concern once more and consider using metrics like cross-validation to observe model performance on unseen data.


Urbanicity of Tracts
Number of Census Tracts
200
100
0
Rural
Urban
Urban Indicator


Housing Units Receiving SNAP in a tract
Number of Census Tracts
40
30
20
10
0
0
250
500
750
Number of Housing Units








```
 PovertyRate      TractSNAP       TractHUNV     MedianFamilyIncome
Min.   : 0.70   Min.   :  0.0   Min.   :  0.0   Min.   : 17404
1st Qu.: 8.25   1st Qu.: 92.5   1st Qu.: 34.0   1st Qu.: 48536
```

```
 Median :14.30   Median :188.0   Median : 80.0   Median : 61579
 Mean   :16.35   Mean   :238.3   Mean   :106.2   Mean   : 68547
 3rd Qu.:21.25   3rd Qu.:347.5   3rd Qu.:145.0   3rd Qu.: 81984
 Max.   :64.20   Max.   :934.0   Max.   :580.0   Max.   :186310
```

```
# A tibble: 2 x 3
  Urban     n  prop
  <fct> <int> <dbl>
1 Rural   164 0.381
2 Urban   267 0.619
```

The univariate EDA for predictors shows how each of the numerical predictors (`PovertyRate`, `TractSNAP`, `TractHUNV`, `MedianFamilyIncome`) have a unimodal, right-skewed distribution. `Urban` shows a similar pattern of positive skew as the majority of census tracts in North Carolina (around 62%) are classified as urban. Due to this skew, the median is a good indicator of the center of the distribution for each of the variables. Thus, we see that the median poverty rate in NC is about 14.30%, which is higher than the United States' official 10.6% poverty rate. The spread of this distribution is also substantially wide in context as the IQR is 13. There are also some outlying census tracts with poverty rates above 50%.

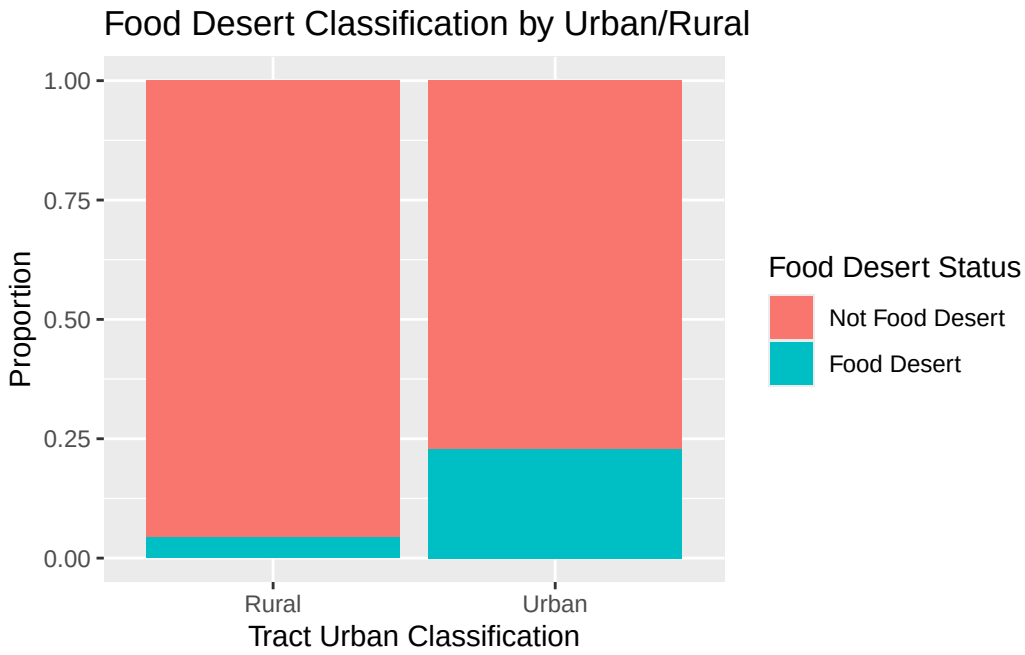
The distribution for `TractSNAP` shows a center around 188, meaning 188 housing units in a census tract receive SNAP benefits. The IQR of 255 housing units also seems relatively large, but when considering the fact that a census tract can range anywhere from 1,200 people to 8,000, this spread seems understandable due to population dynamics and urban/rural distributions. TractSNAP also has outliers at census tracts with more than 400 housing units receiving SNAP benefits.

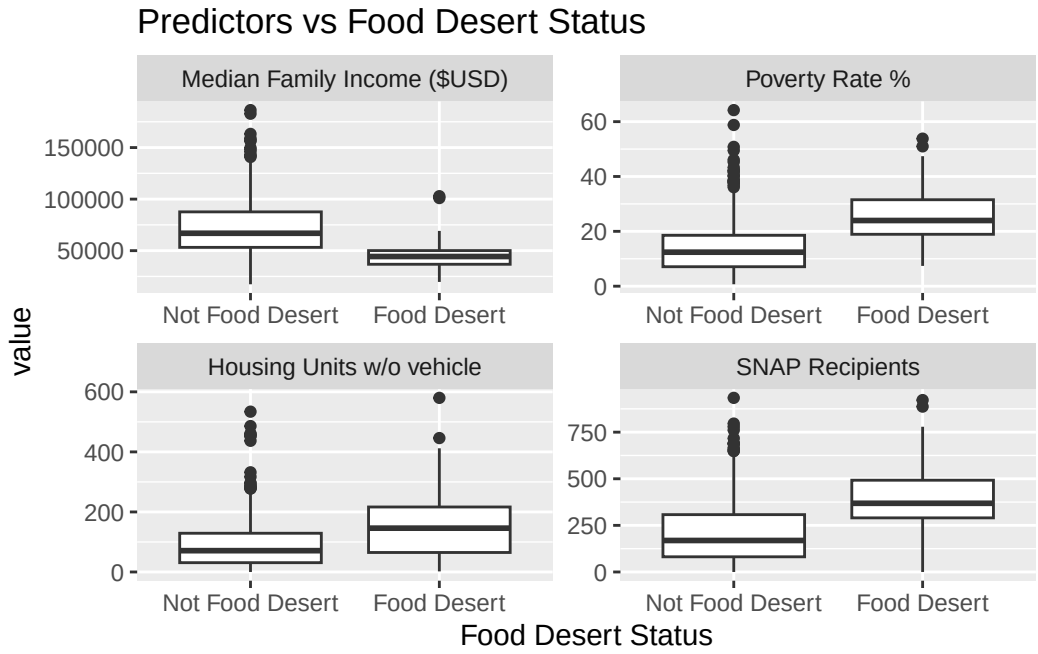
`TractHUNV` has a center of around 80 housing units in a census tract without access to a vehicle. This is surprisingly low but realistic since a car is a necessity in most parts of NC due to the state's lack of public transportation infrastructure. Similar to before, there are a few outliers observed above 400 housing units in a tract without access to a vehicle. The relatively smaller spread (IQR=11) is in line with our understanding of NC's public infrastructure and the necessity of a vehicle, regardless of income status.

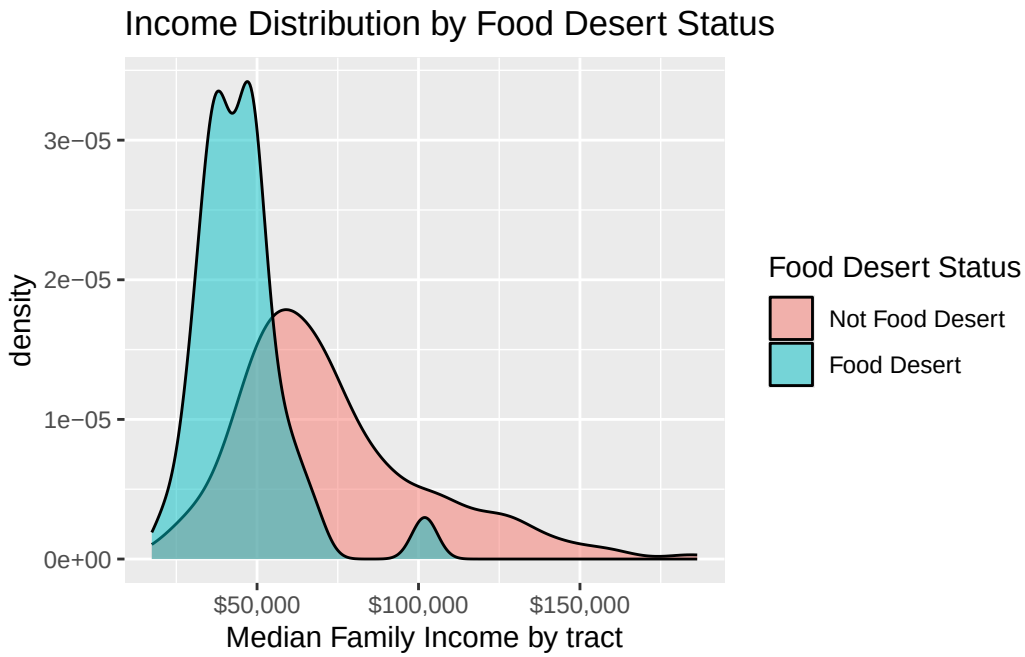
Finally, `MedianFamilyIncome` has a distribution very similar to `PovertyRate` , which is not unexpected since both are roughly measuring the economics state of a census tract. There is a center around $61,500 median family income per census tract, much lower than the US's$83,000 median household income. The spread of the distribution (IQR) is about $20,000, with outliers above median family incomes of $150,000.

Since each of these exhibits a substantial right skew and a couple of outliers, I will use metrics like Cook's distance to decide if points are influential and are harming the model's predictive capabilities. Furthermore, I will check whether these predictors need to be log transformed to represent a nonlinear relationship and account for the skew by comparing the adjusted R^2 of models with different predictors transformed.

Bivariate EDA





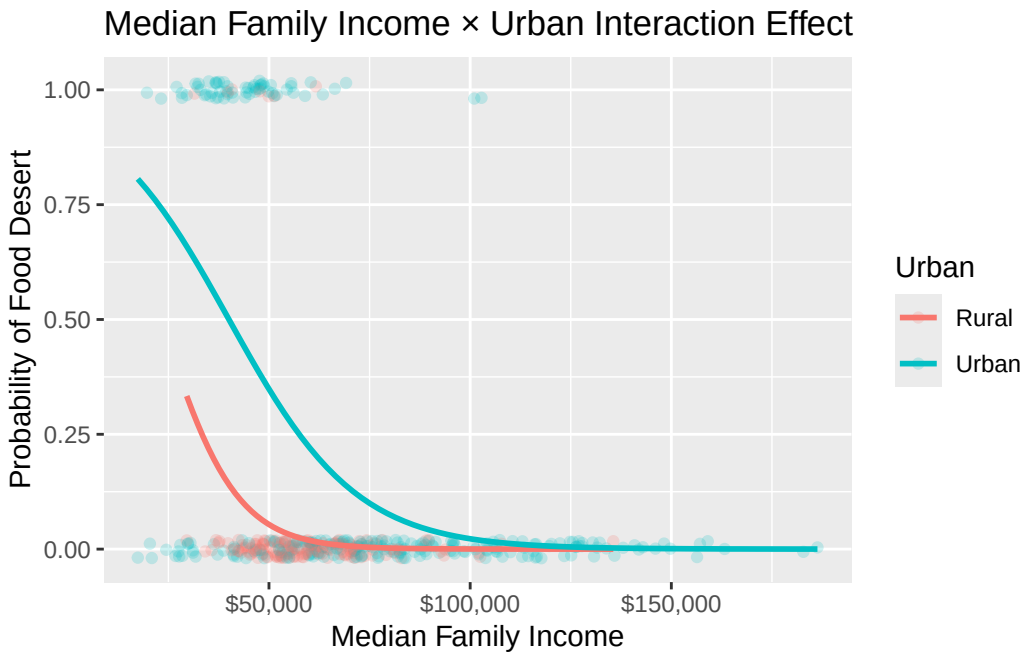


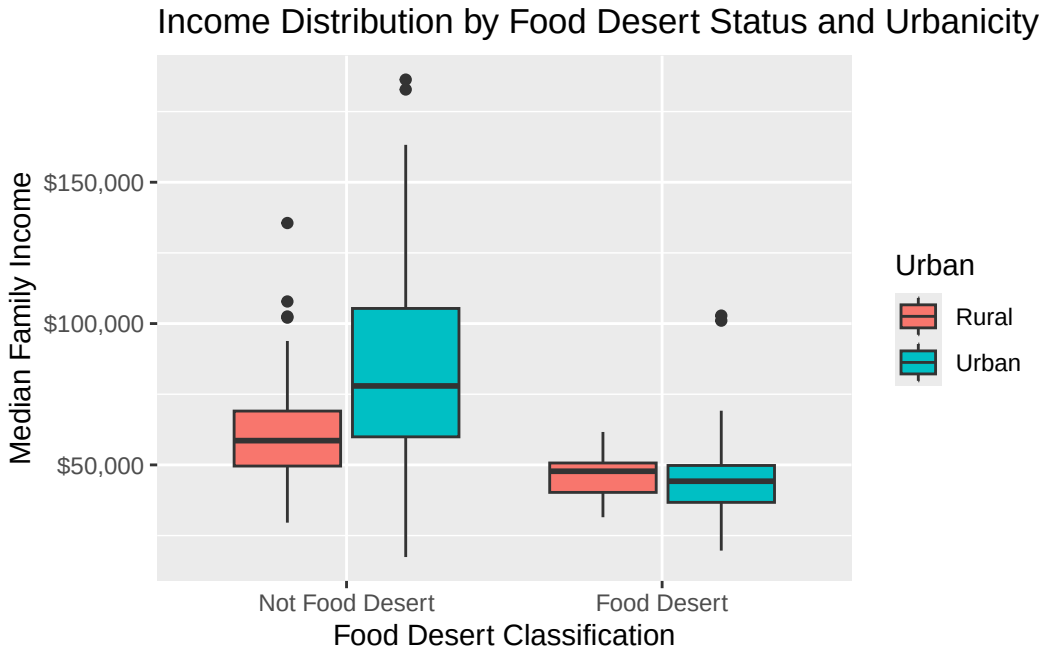
The first plot illuminates an interesting pattern that a higher proportion of urban tracts are food deserts rather than rural tracts. Further analysis in the next steps of this project will investigate specific cases that may contribute to this counterintuitive pattern, as well as explain some hotspots of food deserts observed in the preliminary map. In addition to the facet plot, the double density plot of median family income differing by food desert status is an important visualization to understand how much the distribution of incomes within a tract varies according to whether the tract is in a food desert or not. For instance, tracts that are not food deserts have a similar center to tracts that are food deserts, but the median incomes are more spread out from \$45,000 to \$125,000, while median incomes for tracts in food deserts are mostly concentrated around \$45,000 to \$50,000. This pattern is important for solidifying the necessity of using median family income as a predictor for food desert status. The visualization also promotes an understanding of more routine or customary median family incomes across the state, which is important for conceptualizing the extent to which financial status impacts living conditions and location.

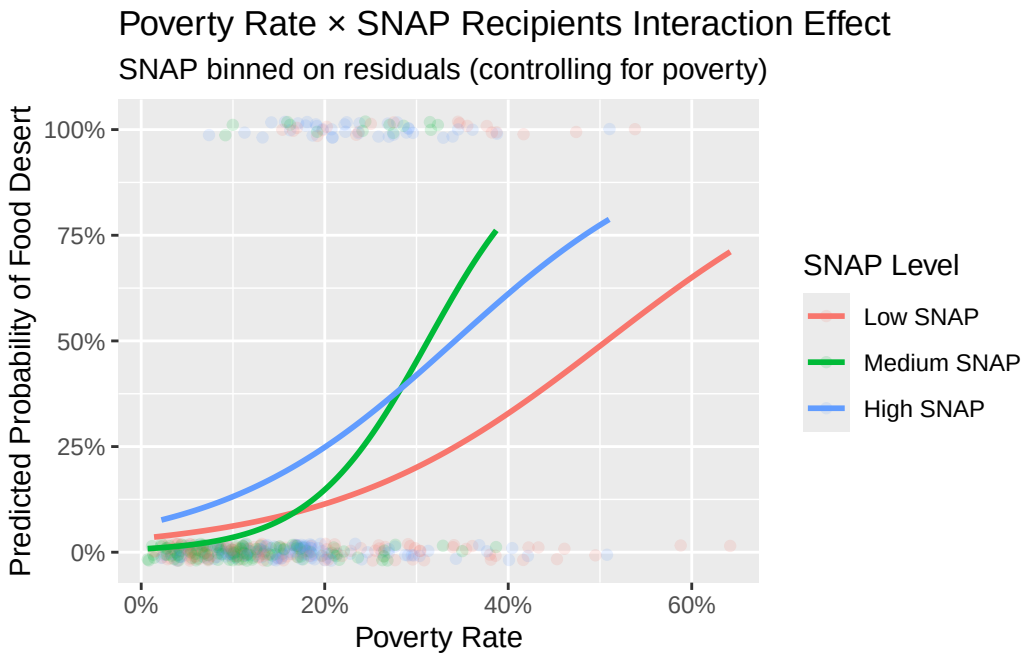
Finally, the facet plot shows very interesting and potentially significantly different relationships for each predictor against food desert status since the boxes of the box plots are (mostly) not contained within each other and vary vertically. This pattern is especially evident in `MedianFamilyIncome`, `TractSNAP`, and `TractHUNV`. As mentioned earlier, tracts that are not food deserts have a noticeably higher median family income, and tracts that are food deserts have both smaller median family incomes as well as smaller variance. The remainder of the predictors show similar patterns to each other, where tracts that are food deserts see a greater number of housing units without a vehicle, a greater number of households receiving SNAP, and higher poverty rates. Each of these see significant differences between their non-food-

desert-census-tract counterparts, but this significance will be reevaluated by assessing p-values of the predictors in the logistic regression model. In addition to providing motivation for using these predictors in the model, the box plots also provide a clearer view of the outliers. Across each of the faceted plots, there seem to be more outliers for tracts that are not classified as food deserts. This may be due to a wider spread of values and less uniformity in socioeconomic characteristics among non–food desert tracts. When creating our model, this knowledge can be helpful in determining other predictors to add or potential interactions with demographic characteristics present in the dataset. As a whole, this depiction of outliers also solidifies the need to assess how influential the outliers are for the model, which will determine if we will remove them for improved predictive power.

Interaction Effects







In the first plot, we see how higher levels of median family income are linked to higher probabilities of food deserts — in both the lines plotted, regardless of urban classification. However, the plot shows that an interaction between urbanicity and median family income certainly exists because the slopes for the two lines are substantially different. The plot shows

that tracts classified as urban have higher probabilities of being food deserts than tracts classified as rural at the same median income. While the lines converge at median family incomes above $125,000, the initial difference serves as evidence that some of the variability between the probability of a food desert and median family income can be explained by a census tract being urban or rural. The second plot serves to reinforce this assessment. The two groups of boxes for tracts classified as food deserts and those as not are significantly different because the boxes do not overlap across groups. The interaction effect between median family income and urbanicity is especially apparent in the census tracts not classified as food deserts, as the plot shows how rural tracts in non-food deserts have a considerably lower median family income than urban tracts. Thus, rather than simply seeing how median family income varies according to food desert classification, adding urbanicity creates an additional, important interaction to consider.

Methodology

```
MedianFamilyIncome         UrbanUrban          TractSNAP   log1p(TractHUNV)
          2.945233           1.096109           1.861732           1.894113
       PovertyRate
          3.065295

   MedianFamilyIncome            UrbanUrban             TractSNAP
             2.895907              1.111105              6.422320
          PovertyRate      log1p(TractHUNV) TractSNAP:PovertyRate
             7.358673              1.917241             11.411684

Analysis of Deviance Table

Model 1: LILATracts_1And10 ~ MedianFamilyIncome + Urban + TractSNAP +
    log1p(TractHUNV) + PovertyRate
Model 2: LILATracts_1And10 ~ MedianFamilyIncome * Urban + TractSNAP +
    log1p(TractHUNV) + PovertyRate
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1       425     245.96
2       424     245.47  1  0.48691   0.4853

# A tibble: 0 x 12
# i 12 variables: LILATracts_1And10 <fct>, MedianFamilyIncome <dbl>,
#   Urban <fct>, TractSNAP <dbl>, PovertyRate <dbl>, log1p(TractHUNV) <dbl>,
#   .fitted <dbl>, .resid <dbl>, .hat <dbl>, .sigma <dbl>, .cooksd <dbl>,
#   .std.resid <dbl>
```

```
# A tibble: 1 x 3
  .metric .estimator .estimate
  <chr>   <chr>          <dbl>
1 roc_auc binary         0.892
```

```
function (object, ..., k = 2)
UseMethod("AIC")
<bytecode: 0x59f35121d118>
<environment: namespace:stats>
```

```
[1] 282.3539
```

```
[1] 255.9343
```

```
[1] 284.3971
```

We will fit a logistic regression model to predict what factors contribute to food deserts across counties in North Carolina. This model is best suited for my dataset because I wish to predict whether a specific census tract is or is not classified as a food desert. Since this is represented as a probability of being a food desert, logistic regression will help make such prediction based on various traditional predictors of food deserts, such as poverty rates, incomes (with SNAP benefits as an additional metric need aware households), vehicle access, and location (urban/rural).

A drop in deviance test comparing two models — one with an interaction effect between median family income and urbanicity and one without — was not statistically significant. This indicates that the interaction term does not significantly improve model fit. However, computing AIC and BIC showed that the interaction model is slightly preferred. Since my main goal with this project is to build a model that predicts well, I will prioritize the AIC and BIC scores and choose the interaction model, containing predictors `PovertyRate`, `TractSNAP`, `TractHUNV`, and an interaction between `MedianFamilyIncome` and `Urban` .

In addition to including an interaction effect in the model, I have also log transformed `TractHUNV` because the EDA showed the distribution to be unimodal with a strong right skew. Since some values of `TractHUNV` are 0, I used log1p() to keep values=0 at 0 since log(0) is undefined. Adjusting for the skew could illuminate nonlinear relationships helpful in predicting the likelihood of food desert presence. Though the histogram for `TractSNAP` was also right-skewed, I did not transform the variable because its bimodal distribution and some data points concentrated even at the wings of the graph suggested that capturing the pattern of the data in its entirety would be important for model prediction. Similarly, since some of the variation in `MedianFamilyIncome` could be explained by `Urban` , I included the interaction term and did not log transform the variable. I did not transform `PovertyRate` because there was not

an extremely strong right skew that necessitated a transformation. Additionally, since I am prioritizing interpretability, log transforming a percent would not be as understandable.

Finally, I computed Cook's Distance and confirmed that the data does not present and influential outliers that could impact the model's fit. Ultimately, after each of these assessments, I decided to use a logistic regression model with `LILATracts_1AND10` as the response variable and `MedianFamilyIncome x Urban` , `TractSNAP` , `log1p(TractHUNV)` , and `PovertyRate` as the predictors.

Results

In this section, you will present the final model and include a brief discussion of the model assumptions, diagnostics, and any relevant model fit statistics.

This section also includes initial interpretations of key model coefficients and discussion of inferential conclusions drawn from the model. The goal is not to interpret every single variable in the model but rather to show that you are proficient in using the model output to address the research question, using the interpretations to support your conclusions. Focus on the variables that help you answer the research question and that provide relevant context for the reader.

do cross validation later to see how well model generalizes to unseen data

term	estimate	std.error	statistic	p.value
(Intercept)	2.126	1.675	0.450	0.653
MedianFamilyIncome	1.000	0.000	-4.144	0.000
UrbanUrban	13.318	0.459	5.644	0.000
TractSNAP	1.007	0.002	3.096	0.002
PovertyRate	1.014	0.037	0.376	0.707
log1p(TractHUNV)	0.730	0.208	-1.509	0.131
TractSNAP:PovertyRate	1.000	0.000	-2.004	0.045

Model assumptions and diagnostics

Logistic regression requires that observations are independent, that the log-odds of the outcome have a linear relationship with continuous predictors, and that there is no severe multicollinearity among predictors. The independence assumption is partially addressed by our random 20% sample of North Carolina census tracts, which reduces (though does not eliminate) the spatial dependence among neighboring tracts. Linearity in the log-odds was assessed through the EDA and transformation of `TractHUNV` to address its strong right skew. Regarding multicollinearity, the VIF scores for the additive model were all below about 3, indicating no problematic collinearity among the predictors. The inflated VIFs in the interaction model

(MedianFamilyIncome: 20.97, Urban1: 25.31, interaction term: 45.16) are expected to be results of including an interaction term rather than evidence of true multicollinearity. Finally, Cook's distance confirmed no influential outliers that would affect model fit.

Model Fit

The interaction model was selected over the additive model based on AIC and BIC, despite a non-significant drop-in-deviance test ($p = 0.485$).The lower AIC suggests the interaction model should predict slightly better, and since the income $\times$ urbanicity interaction makes theoretical sense, supported by the data exploration, it is reasonable to keep it.

Interpretation

While the per-dollar odds ratio appears negligible for median family income, the negative z-statistic confirms that higher income is associated with lower odds of food desert classification. Thus, though the coefficient interpretation is negligible (for each one-unit increase in median family income, the odds that the tract is a food desert are expected to change by approximately 0% on average, holding all else constant), there is at least slight indication by the z statistic that economic resources can be a structural predictor of food access.

TractSNAP is also statistically significant, indicating that for each additional household receiving SNAP benefits in a tract, the odds that the tract is a food desert are expected to increase by 0.3% on average, holding all else constant. This positive association suggests that higher SNAP participation is linked to greater food access limitations, showing that financial assistance alone may not fully address barriers to accessing healthy food.

`PovertyRate` is potentially significant with a p-value of 0.065 in the model and worth investigating. Interestingly, the test statistic is negative, suggesting that an increase in poverty rate is associated with lower odds of a food desert. This pattern contradicts much of the literature on food access and food deserts, so a potential explanation for this direction of change could be collinearity in the model. Regardless, this example demonstrates the importance of interpreting results in context as well as verifying with the literature some examined patterns.

Conclusions

Altogether, the model suggests that economic measures, such as median family income and SNAP benefit receipt, are the most consistent predictors of food desert classification among North Carolina census tracts. The role of the other predictors is more complex, since the higher p-values do not allow us to comment on the significance of those predictors and patterns they illuminate about food access. Cross-validation will be used to further evaluate the model's predictive performance on unseen data, particularly given the class imbalance of 84% of census tracts being categorized as not a food desert.